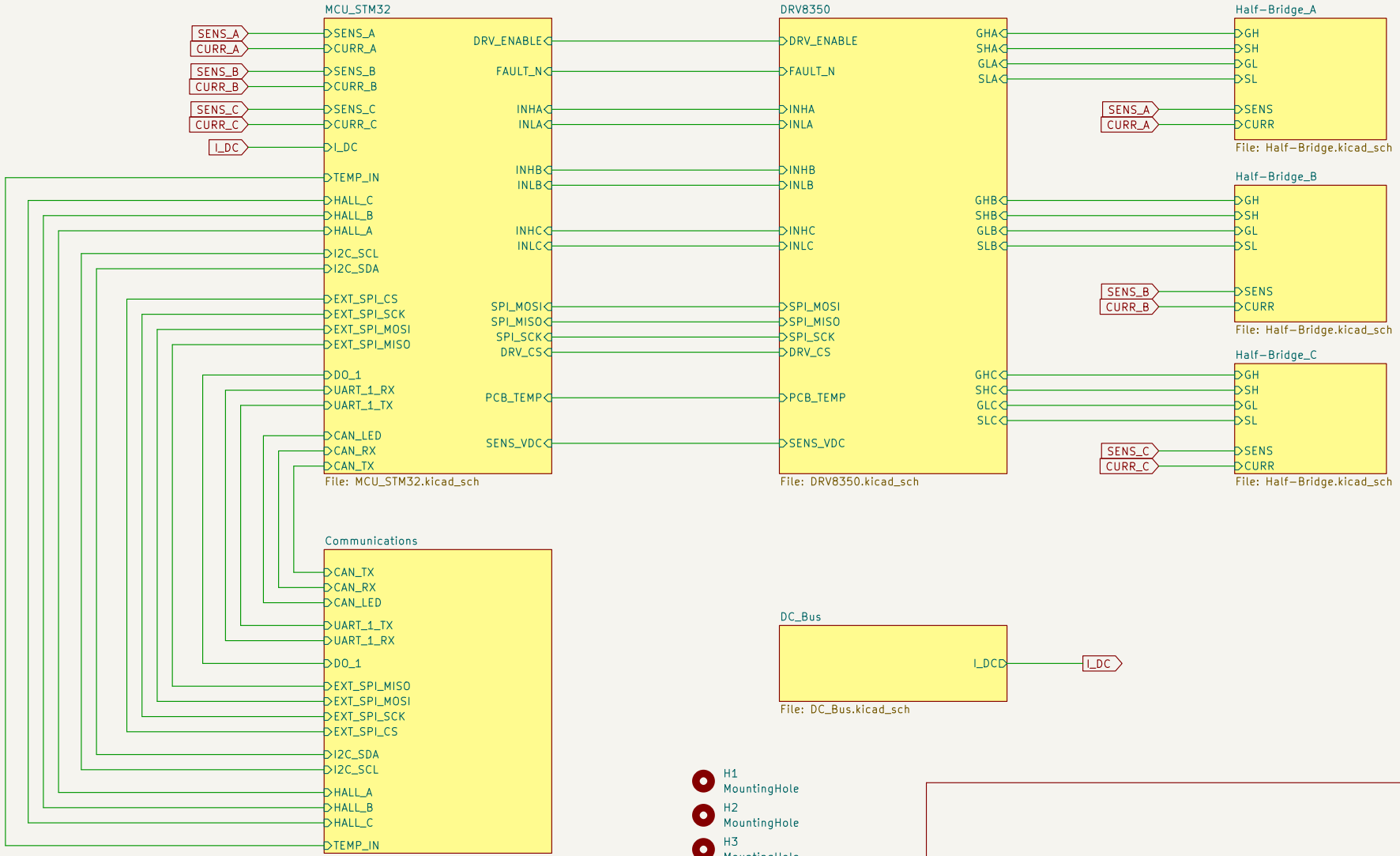
@Todo List:
1- How to import PCB layer stack up
2- Add 1x10uF, 2x100nF X7R ceramic (Place near MOSFETS)



Sheet: /
File: BLDC_V_0.1.kicad_sch

Title:

Size: A4	Date:	**Rev:**
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MCU

U1A

STM32G473VET6

I_DC	20	PA0	
CURR_A	21	PA1	
	22	PA2	
CURR_B	25	PA3	
EXT_SPI_CS	26	PA4	
EXT_SPI_SCK	27	PA5	
EXT_SPI_MISO	28	PA6	
EXT_SPI_MOSI	29	PA7	
SENS_VDC	69	PA8	
	70	PA9	
	71	PA10	
USB_D-	72	PA11	
USB_D+	73	PA12	
SWDIO	76	PA13	
SWCLK	77	PA14	
I2C_SCL	78	PA15	
TEMP_IN	15	PC0	
	16	PC1	
	17	PC2	
PCB_TEMP	18	PC3	
UART_1_RX	30	PC4	
UART_1_TX	31	PC5	
INHA	65	PC6	
INHB	66	PC7	
INHC	67	PC8	
	68	PC9	
INLA	79	PC10	
INLB	80	PC11	
INLC	81	PC12	
	7	PC13	
	8	PC14-OSC32_IN	
	97	PE0	
PROCCESS_LED	98	PE1	
HALL_A	1	PE2	
HALL_B	2	PE3	
HALL_C	3	PE4	
CAN_LED	4	PE5	
	5	PE6	
CURR_C	38	PE7	
	39	PE8	
	40	PE9	
	41	PE10	
	42	PE11	
	43	PE12	
SENS_C	44	PE13	
SENS_A	45	PE14	
SENS_B	46	PE15	
	9	PC15-OSC32_OUT	
HSE_OUT	13	PF1-OSC_OUT	
DO_1	32	PB0	
	33	PB1	
	34	PB2	
SWO	90	PB3	
	91	PB4	
	92	PB5	
	93	PB6	
FAULT_N	94	PB7	
BOOT0	95	PB8-BOOT0	
I2C_SDA	96	PB9	
	47	PB10	
	50	PB11	
DRV_CS	51	PB12	
SPI_SCK	52	PB13	
SPI_MISO	53	PB14	
SPI_MOSI	54	PB15	
CAN_RX	82	PD0	
CAN_TX	83	PD1	
HALL_C	84	PD2	
	85	PD3	
	86	PD4	
	87	PD5	
	88	PD6	
	89	PD7	
	55	PD8	
	56	PD9	
	57	PD10	
	58	PD11	
	59	PD12	
	60	PD13	
	61	PD14	
DRV_ENABLE	62	PD15	
HSE_IN	12	PF0-OSC_IN	
	19	PF2	
	10	PF9	
	11	PF10	
NRST	14	PG10-NRST	

HSE_IN — Y1 24MHz — R1 47R — HSE_OUT

C1 20p — GND

C2 20p — GND

PROCCESS_LED — D1 BLUE — R2 1k5 — GND

MCU Power

U1B

STM32G473VET6

+3.3V	6 VBAT
	24 VDD
	49 VDD
	64 VDD
	75 VDD
	100 VDD
+3.3VA	37 VDDA
	36 VREF+
GND	23 VSS
	48 VSS
	63 VSS
	74 VSS
	99 VSS
	35 VSSA

Decoupling caps should be close as possibile to the mcu

+3.3V

C3 100n — C5 100n — C6 100n — C7 100n — C9 100n — C11 100n — C12 100n — C15 100n — C18 10u

GND

LDO_IN — U2 AMS1117-3.3 (VI, VO, GND) — +3.3V

C4 4u7 — C8 22u — GND

BLM18PG221SN1D — L1 — PWR_FLAG — +3.3VA

D3 RED — R4 1k — C13 1u — C16 100n — GND

LDO_IN — PWR_FLAG — +5V — PWR_FLAG — D2 BAT54C — VBUS

NRST — C10 100n — GND

+3.3VA — C14 100n — C17 10u — GND

Program Pins and USB

+3.3V — J2 SWD — SWDIO, SWCLK, SWO, NRST — GND

BOOT0 — R3 10k — +3.3V — J3 Conn_01x03 — GND

VBUS — U3 USBLC6-2SC6 — USB_D- — USB_D+ — USB_CONN_D- — USB_CONN_D+ — GND

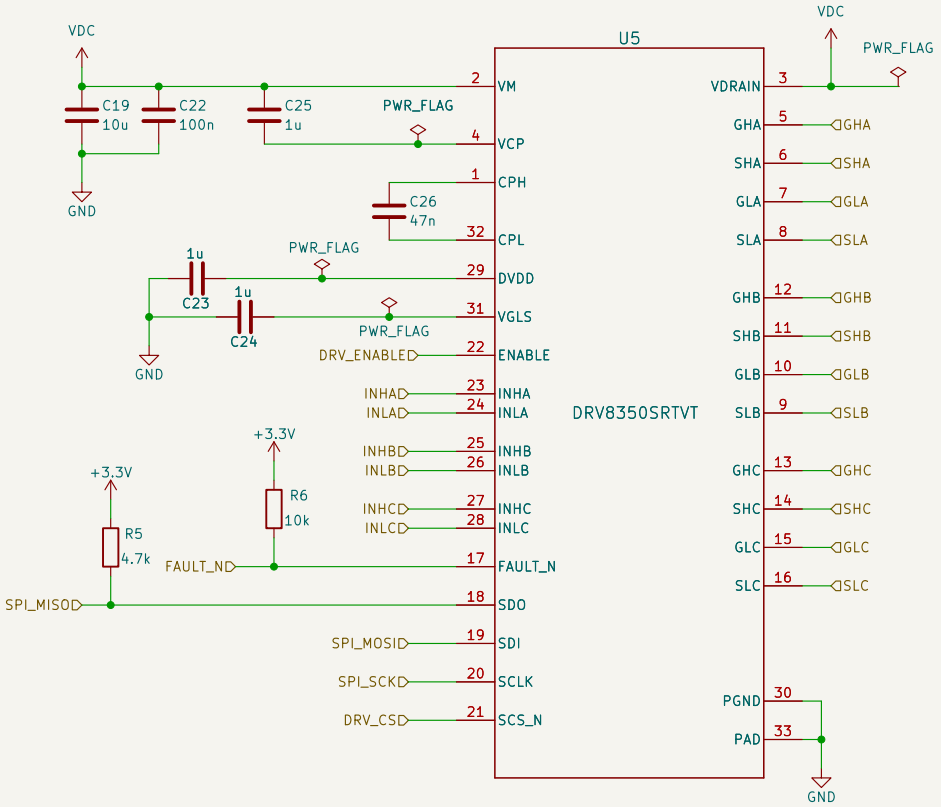
J1 USB_B_Micro — VBUS — D+ — D- — ID — GND — Shield — SH — USB_CONN_D+ — USB_CONN_D- — VBUS — GND

Sheet: /MCU_STM32/
File: MCU_STM32.kicad_sch

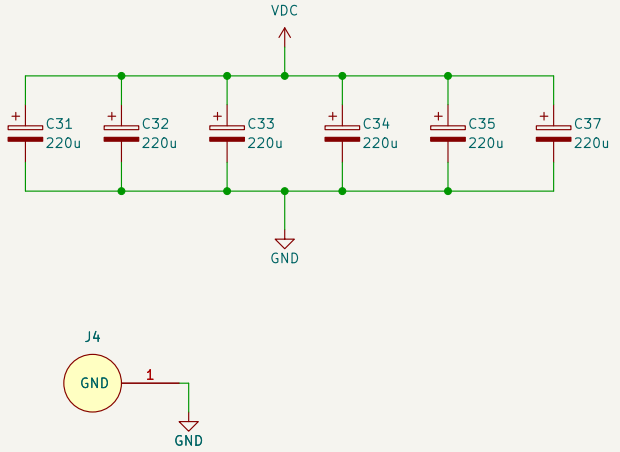
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Size: A4 | Date: | **Rev:**
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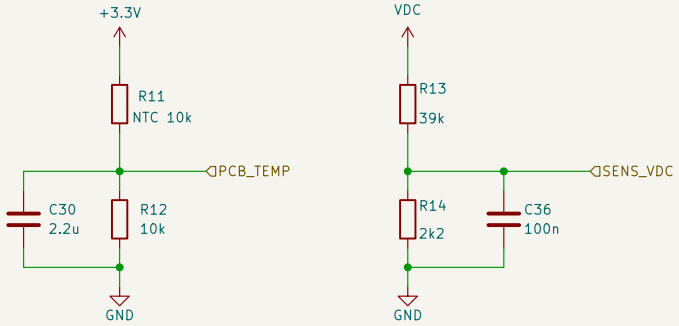
Gate Driver



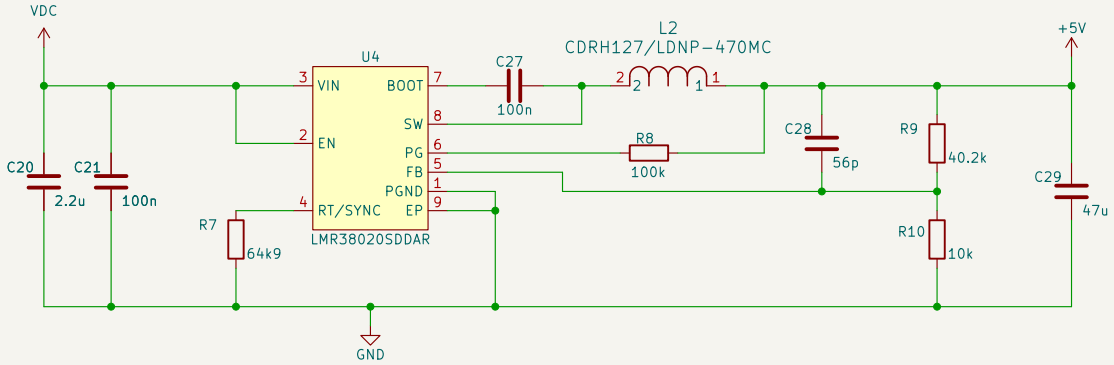
Bulk Capacitors



Temperature and Voltage Measurements



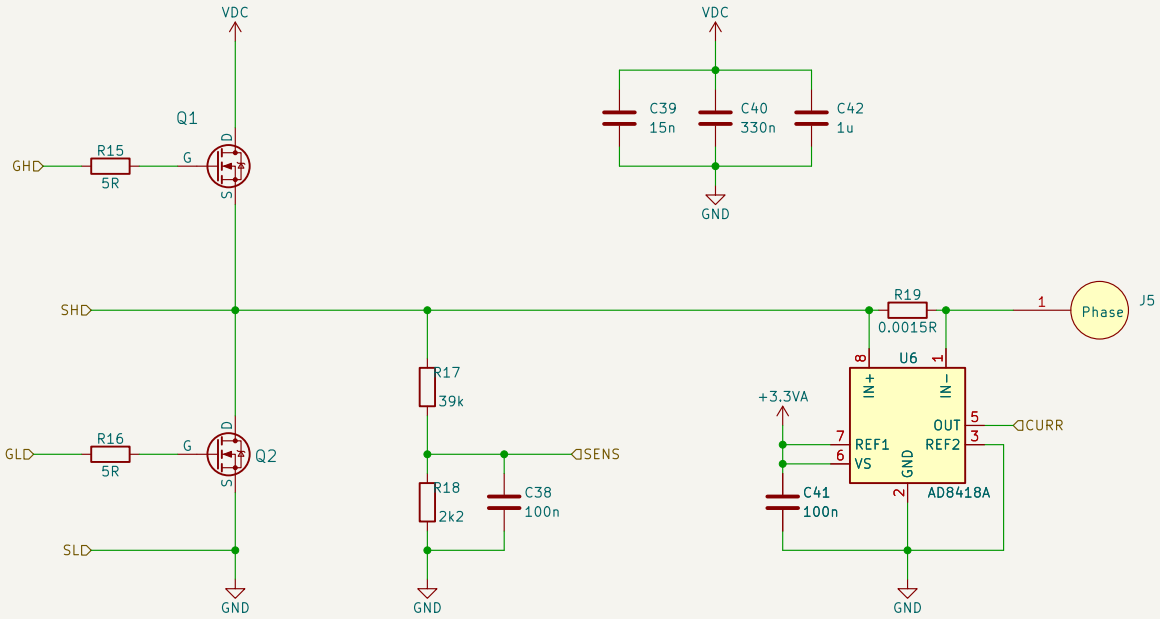
Buck Converter



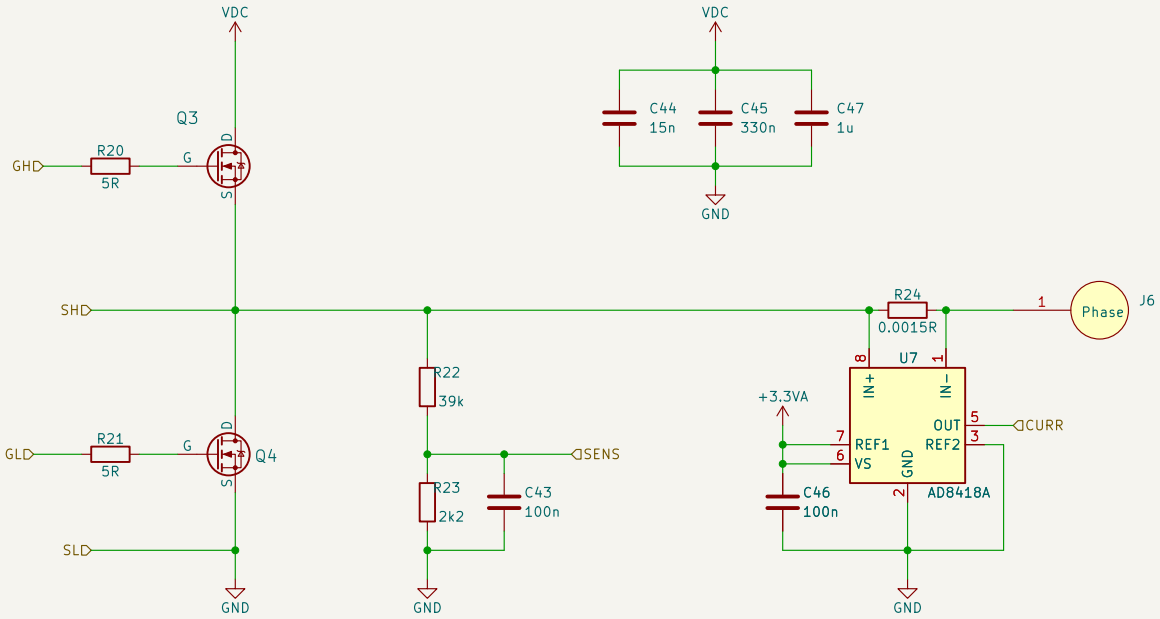
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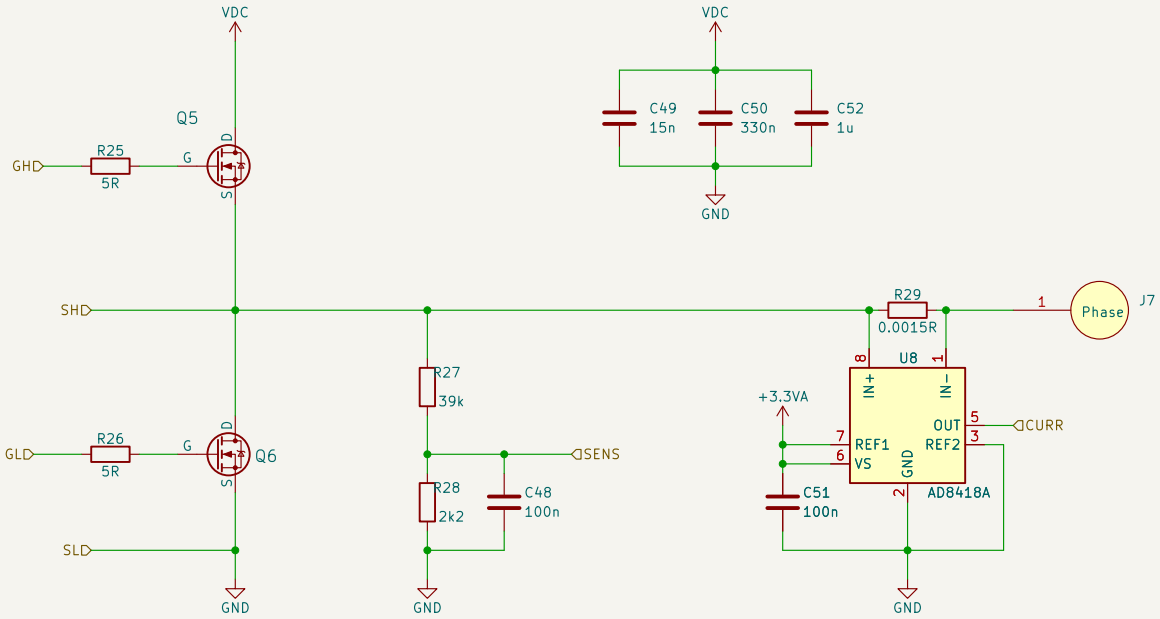
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 3/8



Sheet: /Half-Bridge_A/		
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Title:		
Size: A4	Date:	**Rev:**
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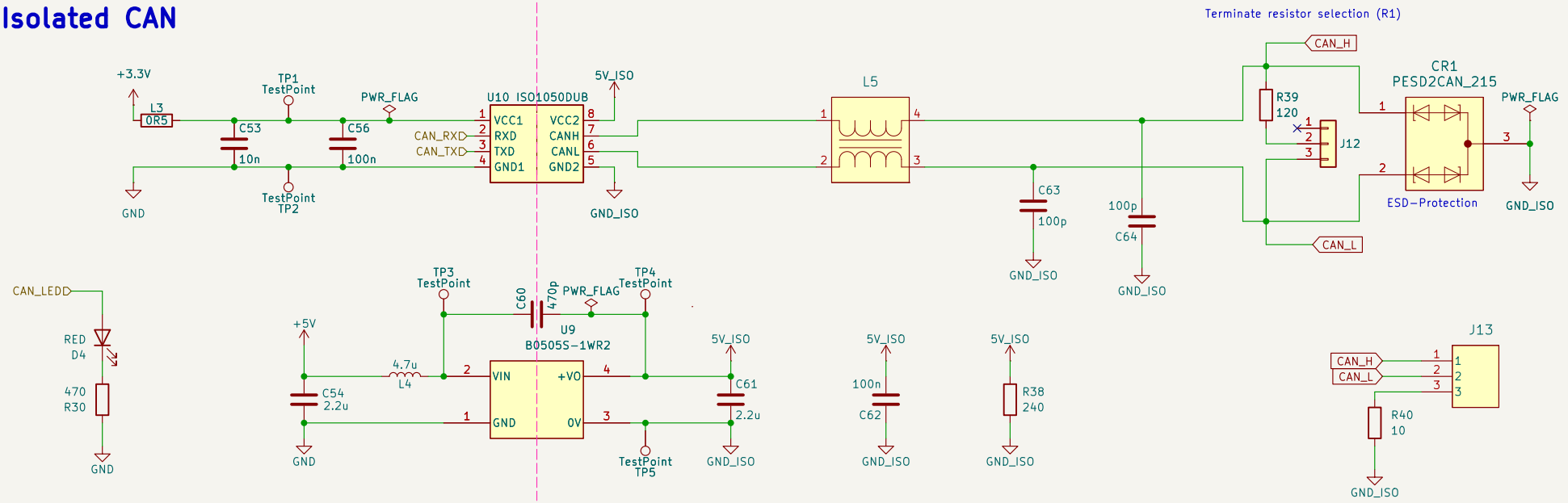


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KiCad E.D.A. 10.0.6		Id: 5/8

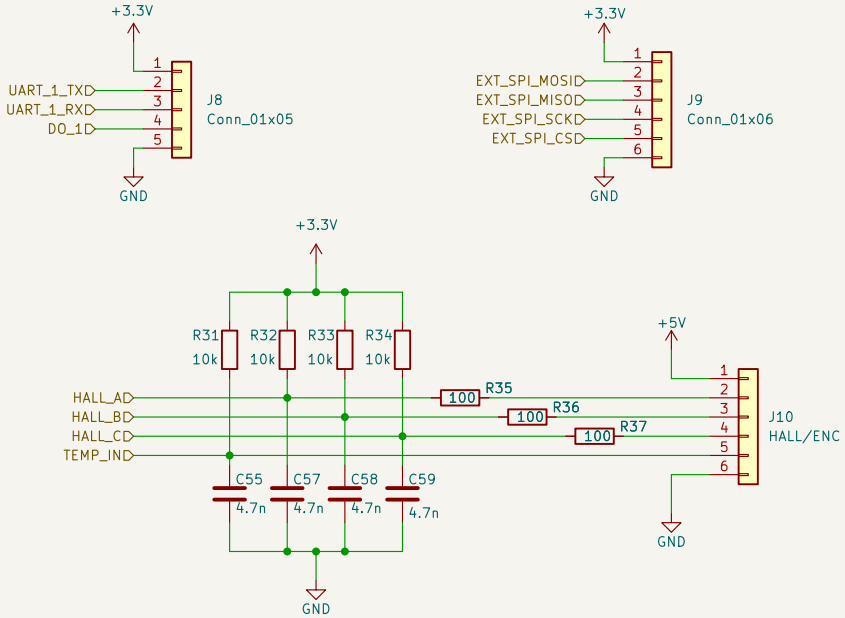


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KiCad E.D.A. 10.0.6		Id: 6/8

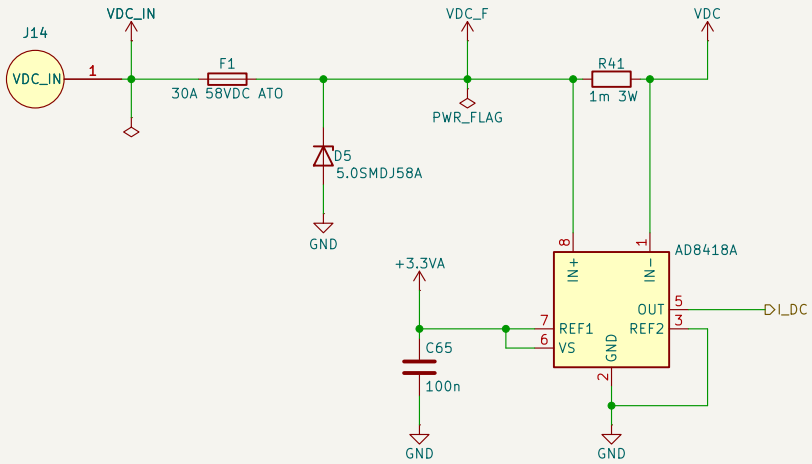
Isolated CAN



Connectors



Sheet: /Communications/		
File: Communications.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
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Sheet: /DC_Bus/		
File: DC_Bus.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 8/8